ALBUKHARY INTERNATIONAL UNIVERSITY

SCHOOL OF COMPUTING AND INFORMATICS

SEMESTER 2

2024/2025 CCE2233

Requirements Engineering

GROUP PROJECT

PROJECT WEIGHT: 20%

LECTURER

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TITLE

Flight Reservation System

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1.Executive Summary

The present project is dedicated to the creation of a complete and ethically sound requirements specification of a Flight Reservation System aiming at increasing the efficiency, security, and user experience of booking flights. The increasing use of online booking systems has increased the need to have systems that should be fast, secure, user-friendly, and should be able to meet the needs of various stakeholders involved in these solutions such as the passengers, the travel agents, airline employees, and the system administrator.

A systematic manner of engineering requirements followed, considering that the project was using stakeholders surveys to come up with functional, as well as non-functional requirements. The most essential features are a real-time flight search, seat selection, group booking, the potential to make safe online payments, making personalized recommendations, and his or her dashboards. The non-functional requirements that were prioritized included the system speed, availability, scalability and data security in order to guarantee a good system performance and reliability.

Ethics was incorporated in all aspects of the project life cycle, focusing on the fair, transparent, stakeholder privacy and giving informed consent. Reviewed some of the possible ethical issues like the privacy of the data and unreasonable user requirements and listed the possible solutions to the challenges. The project ended with an organized, traceable and prioritized requirements package and therefore the system is sure to achieve all its operational objectives as well as the expectations of the stakeholders without any breakdown in ethics.

3. Introduction

3.1 Project Context

The process of flight reservation has been undertaken via the flight reservation systems which have brought much needed ease and convenience within the contemporary travel sector so that passengers, travel agents, as well as airline employees can coordinate and make flight reservations less stressful. Systems of Airline Reservations is a system that is computerized and helps in retrieving information thereby conducting transactions connected to the travel of air. These systems have been designed originally by the airlines and have been extended at a later stage for travel agencies' usage (Antor et al. ,2021). As more people turn towards online services, these systems must be quick, safe, and convenient with the possibility to include seat selection, payment system, and booking system. The main idea behind our project is to analyze and specify the requirements to a robust flight reservation system that should support the needs of different stakeholders of such a system, such as passengers, agents, and staff of airlines.

3.2 Objectives and Scope

The primary objective of this project is to gather, document, and manage both functional and non-functional requirements for a Flight Reservation System using ethical practices. The scope includes identifying user expectations through surveys, documenting core features like real-time flight search and booking management, and ensuring that privacy, security, and fairness are integrated throughout the requirements engineering process. The system is designed for end-users such as passengers and agents who frequently or occasionally book flights online.

4. Methodology

4.1 Elicitation Techniques Used

A designed Google Forms survey on 33 respondents of different backgrounds was completed to collect accurate and stakeholder-centered requirements. The participants were interviewed regarding their place in flight booking (e.g., a passenger, an agent), their booking rate, and their expectations about features. There were questions with multiple choice and Likert answers, addressing both functional (e.g., making-a-booking, payment, seat selection) and non-functional (e.g., speed of the system, ease of use, security) functions. This method spared a wide coverage of user requirements as well as preferences. The survey responses were quantitatively analysed to determine the capabilities of the system requested most.

4.2 Ethical Considerations in Stakeholder Engagement:

The survey process adhered strictly to ethical standards:

- Informed Consent: Participants were clearly informed that the survey was voluntary and their data would only be used for academic purposes.

- Privacy Protection: No sensitive personal information (e.g., passport numbers, financial data) was collected. Names were collected only to verify genuine participation but were not used for analysis.

- Data Handling: The result we obtained in the survey remained confidential and used strictly for requirement analysis. Any identifiable data was anonymized during report preparation.

- Transparency and Fairness: All participants had equal opportunity to voice their expectations. No response was prioritized based on identity or background.

- Consent for Storage and Use: Stakeholders were asked about their comfort level with data being stored for future bookings, and their responses guided our approach to security requirements.

5. Requirements Specification

5.1 Elicited Requirements

Based on the survey and stakeholder roles (Passenger, Travel Agent, Airline Staff, System Administrator), the following requirements were identified:

- The system shall allow all users to register and log in to their accounts.
- The system shall redirect users to their respective dashboards (passenger, travel agent, staff, admin) after login.
- The system shall allow passengers and agents to search for flights by date and destination.
- The system shall provide filtering and sorting options (price, airline, schedule).
- The system shall support seat selection prior to booking.
- The system shall allow group bookings.
- The system shall allow passengers and agents to make secure online payments.
- The system shall send booking updates via email, SMS, app notifications, or calls.
- The system shall allow airline staff to manage flights, seats, and prices.
- The system shall notify staff in real time about bookings or cancellations.
- The system shall allow passengers to view booking history and track ticket status.
- The system shall store user preferences and provide personalized recommendations.
- The system shall provide multi-language support.
- The system shall run 24/7 with minimal downtime.
- The system shall ensure strong data encryption and protect personal/payment data.
- The system shall restrict unauthorized access to admin dashboards.
- The system shall process actions (searching, booking) within 3 seconds.
- The system shall be compatible with all modern browsers (desktop and mobile).

5.2 Functional Requirement

Req ID	Requirement Description	Priority	Source
FR-01	Users shall be able to register and log in to their accounts.	High	All Stakeholders
FR-02	Users shall be redirected to role-based dashboards after login.	High	All Stakeholders
FR-03	Passengers and travel agents can search for flights by date and destination.	High	Passenger, Travel Agent
FR-04	Filtering and sorting flights by price, time, or airline.	Medium	Passenger, Travel Agent
FR-05	Passengers shall be able to select seats.	High	Passenger
FR-06	The system shall support group booking for multiple passengers.	Medium	Passenger, Travel Agent
FR-07	Online payments shall be accepted (cards, ewallet, bank transfer, etc.).	High	Passenger, Travel Agent

FR-08	Notifications shall be sent via email, SMS, app, or phone call.	High	Passenger, Travel Agent
FR-09	Airline staff shall manage flight details, seat availability, and prices.	High	Airline Staff
FR-10	Real-time updates about bookings and cancellations shall be sent to airline staff.	High	Airline Staff
FR-11	Passengers shall be able to view booking history and track flight status.	Medium	Passenger
FR-12	Personalized recommendations shall be based on preferences or past bookings.	Medium	Passenger
FR-13	Admin shall manage user accounts, access rights, and oversee the system.	High	System Administrator
FR-14	The system shall restrict unauthorized access to admin and airline staff dashboards.	High	System Administrator

5.3 Non- Functional Requirement

NFR ID	Requirement Description	Priority	Source
NFR-01	The system shall support all modern web browsers (desktop and mobile).	High	Technical Team
NFR-02	All user actions (search, payment) shall be processed within 3 seconds.	High	Usability Standard
NFR-03	The system shall run 24/7 with minimal downtime.	High	All Stakeholders
NFR-04	User data must be encrypted and securely stored.	High	Privacy Concern (Survey Data)
NFR-05	The system shall support multiple languages for global accessibility.	Medium	Survey Participants
NFR-06	The system shall scale to handle peak booking periods without performance degradation.	Medium	Technical Requirement

5.4 Requirement Prioritization and Rationale

Effective software Requirement Prioritization plays an essential role in the success of the Software Development process, ultimately contributing to the successful delivery of high-quality software. . Among the various methods for Requirement Prioritization, the MoSCoW method has gained widespread adoption due to its ease of use (Vijayakumar et al., 2024). In our project, the project requirements have been prioritized using the MoSCoW method: Must have, Should have, Could have, and Won't have for this release. Prioritization is based on survey response percentages, stakeholder criticality, and technical feasibility.

Req ID	Requirement Description	Priority (MoSCoW)	Rationale

FR-01	Users shall be able to register and log in to their accounts.	Must Have	Core access control. Required by all roles (Passengers, Staff, Admin).
FR-02	Role-based redirection after login.	Must Have	Ensures that users only see relevant dashboards. Supports system modularity.
FR-03	Search flights by date and destination.	Must Have	84.8% of survey respondents expect this feature. Essential for core functionality.
FR-04	Filter/sort flights by price, time, or airline.	Should Have	68.7% of users want filtering and improves usability but not core to basic booking.
FR-05	Passengers shall be able to select seats.	Must Have	81.8% selected this. Critical for user satisfaction and booking completion.
FR-06	Group booking support.	Must Have	84.8% of the users supported this and essential for families or agencies
FR-07	Online payments (cards, e-wallets, bank transfer, etc.).	Must Have	87.9% of respondents expect payment features. Core to transaction completion.
FR-08	Notifications via email	Must Have	90.9% want confirmations, needed to ensure timely and clear communication.
FR-09	Airline staff manage flight details, seats, and prices.	Must Have	Critical backend functionality for airline operations.
FR-10	Real-time updates of bookings/cancellations for airline staff.	Must Have	Needed to synchronize systems and prevent overbooking; supports operational efficiency.
FR-11	Passengers view booking history and flight status.	Should Have	51.5% selected this which is helpful for users, especially repeat travelers.
FR-12	Personalized recommendations.	Could Have	33.3% interest in this which is a good future enhancement but not a launch priority.
FR-13	Admin manages user accounts, roles, and oversees the system.	Must Have	Required for governance and system maintenance.
FR-14	Restrict unauthorized access to admin/staff dashboards.	Must Have	High security priority, protects sensitive data and operations.

NFR ID	Requirement Description	Priority (MoSCoW)	Rationale
NFR-01	Support web browsers and mobile apps (desktop & mobile).	Should Have	Ensures usability across platforms; 51.5% expectation from survey.
NFR-03	Actions (e.g., search, payment) processed within 3 seconds.	Must Have	Most of the user expectation; ensures system responsiveness.
NFR-04	System uptime 24/7 with minimal downtime.	Must Have	97% of respondents indicated this is critical.
NFR-05	Encrypt and securely store user data.	Should Have	60.6% want secure handling or consent before data use.
NFR-06	Support multiple languages.	Could Have	39.4% rated this feature highly; important for global accessibility.
NFR-07	System scalability to handle peak periods.	Should Have	Inferred from technical and usability expectations and importance during holidays/sales.

6.0 Requirement Management

6.1 Traceability Matrix

Req ID	Requirement Description	Source	Status
FR-01	Users shall be able to register and log in to their accounts.	All Stakeholders	Approved
FR-02	Users shall be redirected to role-based dashboards after login.	All Stakeholders	Approved
FR-03	Passengers and travel agents can search for flights.	Passenger, Travel Agent	Approved
FR-04	Filtering and sorting flights.	Passenger, Travel Agent	In Progress
FR-05	Passengers shall be able to select seats.	Passenger	Approved
FR-06	Group booking for multiple passengers.	Passenger, Travel Agent	Proposed
FR-07	Online payments (card, E-wallet, etc.).	Passenger, Travel Agent	Approved
FR-08	Notifications (email, SMS, app, phone call).	Passenger, Travel Agent	In Progress
FR-09	Airline staff manage flights, seats, prices.	Airline Staff	Approved
FR-10	Real-time updates for bookings/cancellations to airline staff.	Airline Staff	Approved
FR-11	Passengers can view booking history & track flight status.	Passenger	In Progress

FR-12	Personalized recommendations.	Passenger	Proposed
FR-13	Admin manages user accounts and access.	System Administrator	Approved
FR-14	Restrict unauthorized access to staff/admin dashboards.	System Administrator	Approved

NFR ID	Requirement Description	Source	Status
NFR-01	The system shall support modern web browsers (desktop & mobile).	Technical Team	Approved
NFR-02	All user actions processed within 3 seconds.	Usability Standard	In Progress
NFR-03	System shall run 24/7 with minimal downtime.	All Stakeholders	Approved
NFR-04	User data encrypted and securely stored.	Survey Participants	Approved
NFR-05	Support for multiple languages.	Survey Participants	Proposed
NFR-06	Scalability for peak booking periods.	Technical Requirement	In Progress

6.2 Change management and versioning

Version	Date	Team	Description of Changes	Status
v1.0	10-June-2025	Requirements Analyst	Initial draft based on survey responses and stakeholder inputs	Draft
v1.1	13-June-2025	Project Team	Added functional, non-functional requirements and stakeholder roles	In Review
v1.2	17-June-2025	Requirements Analyst	Prioritized using MoSCoW, added rationale with survey percentages	Approved
v1.3	20-June-2025	Project Team	Added Traceability Matrix and Change Management process	Approved
v2.0	22-June-2025	Project Team	Finalized baseline for development phase	Finalized

7. Ethical Practices

7.1 Ethical Dilemmas Encountered

In the process of carrying out the survey and collecting system requirements for the Flight Reservation System, several ethical issues emerged that required careful consideration, consistent with the growing need to incorporate ethical thinking in software development. These dilemmas align with the findings in recent research on requirements engineering, which underscores the importance of addressing ethical concerns during the elicitation process (Koopstra, 2020).

7.1.1 Privacy of Survey Respondents

When constructing the user survey, we initially discussed collecting names or email addresses so that we might follow up. However, we were aware that this could generate privacy concerns. In order to address this, we made the survey anonymous and refrained from gathering any personally identifying information (PII). This guaranteed that participants could react freely and without concern for data exploitation.

7.1.2 Fair Prioritization of Requirements

The majority of survey respondents were passengers, which could lead to a bias towards features that are important to passengers. As a result, we balanced functional needs across roles and matched each requirement with its source, ensuring that all stakeholders were appropriately represented, particularly administrators, airline employees, and travel agents.

7.1.3 Including Optional but Ethical Features

Features like multi-language support and special assistance options for passengers with disabilities were not highly prioritized in the survey. However, our team chose to keep them as "Should Have" or "Proposed" to guarantee that the system stays inclusive and ethically responsible, even if such features are not included in the initial release.Even if certain features are not immediately critical, they play an important role in fostering inclusivity, a key ethical consideration in system design(Koopstra, 2020).

7.1.4 Security and Misuse Prevention

We discussed the possibility of unauthorised individuals accessing admin dashboards or automating fraud bookings. To prevent such misuse, we incorporated reliable access controls, encrypted data storage, and real-time system monitoring into the non-functional needs.

7.2 Resolutions and Compliance Strategies

To address the dilemmas and ensure ethical compliance, the following strategies were applied:

- Unambiguous Participation: The participants were given a brief describable description of what the survey will be used for, the permission to the usage of the data, and that they chose to complete the questionnaire before entering into the questionnaire.

- Limited Data Collection: Only necessary (not confidential), non-sensitive data was collected (e.g. user role, booking preferences). There was no request for any money, identity, and secret personal information.

- Anonymity and Confidentiality: The names of the participants were taken so as to ensure that they gave honest answers but it was not counted as part of the data analysis.

Results of the surveys were all anonymized when documenting and reporting.

- Expectation Management: Survey instructions made it clear that not every suggestion can be offered instantly but all feedback would be directly involved in the prioritization of requirements.

- Alignment with Academic Standards: All processing of data and survey behavior were conducted in line with those of a standard academic research with an exclusive purpose of educational and system design application.

7.3 Fairness, Transparency, and Stakeholder Privacy

Throughout the project, fairness, transparency, and privacy were maintained as core principles:

- Fairness: No group of constituents was favored over the other because all the parties such as passengers, agents, staff, and administrators enjoyed equal opportunities to present their opinions.

- Transparency: The nature of the survey and how results would be made available to the participants and the fact that participation was voluntary were clearly stated in advance.

- Protection of the Privacy of Stakeholders: It was not stated in the requirements documentation that identifiable personal data were included. Stakeholder privacy concerns in the survey directly influenced stakeholder requirements of data encryption, data access limitations, and secure data storage.

- Data-Based Requirement Design: Through the use of data, requirements were made based on user expectations and ethical values of protecting privacy and marginalization of users, such as secure login system, encrypted transaction, limited access to administration, and preferences-led personalization.

By following these practices, the Flight Reservation System requirements were gathered responsibly, ensuring stakeholder voices were heard without compromising ethical standards.

8.Reflection

In the process of developing a Flight Reservation System project on our part, a lesson of significant value was learnt by our group in utilizing the theoretical concepts of requirements engineering in a real world system. Each step of defining stakeholders, designing, analysing and defining requirements to the survey allowed us to appreciate the importance of organised and user-friendly strategy to the development of effective systems. At the requirement elicitation phase, one of the most important learning experiences was an attempt to create a survey and solicit input from 33 respondents. Through this we understood how important it is we come up with clear and comprehensible questions and make good analyses of information so that we come out with meaningful results. By means of translating user preferences into quantitative requirements, we ensured that our system is developed based on real demands instead of speculations.

We also discovered the ways of prioritising requirements according to the MoSCoW technique and how to identify functional versus non-functional requirements. Not only did it underline the importance of non-functional aspects such as performance, availability and data security but it also helped us to focus on the most important items such as flight search, seat selection and payment integration. Working on traceability matrix, change management and versioning also allowed us to further understand how to be more consistent and in control of a project, especially when some changing needs arise. It has also taught us how valuable documentation can be towards the maintenance of responsibility and clarity in the development process. Moreover, since we prepared tasks such as survey writing, analysis of requirements, writing and formatting documents, and some activities by splitting them, the team work enabled us to get more efficient in collaboration with each other. Frequent meetings enabled us to bridge the gaps between us namely to fit the stakeholder needs to the technology feasibility as well as the realization of scope with the project deadline.

In general, this project allowed us to narrow a gap between theoretical knowledge and practice. It enforced the notion of the fact that requirement engineering is not only about the list of features, but it is also about the knowledge of the users, change management, the design of trustworthy and ethical systems.

9.Conclusion

In today's time, the world has moved to the advanced way of internet technology, even tasks such as paying the bills, booking vacations, and shopping have become an online business. The airline industry has taken advantage of the rise of the Internet for developing the reservation systems of the airlines(Shaikh & Borate , 2023) In our project, the Flight Reservation System project has been implemented in systematic use of ethical, user-centered and technically stable practices towards collection, analyses and documentation of system requirements. Needs and expectations of various stakeholders (that is, passengers, agents, airline personnel, and administrators) were successfully elicited through the use of well-designed surveys. The order of precedence of functional and non-functional requirements was decided as per the input of the real users so that the design of the system matches the requirements of the operations as well as user satisfaction. Fairness, privacy, and transparency were observed during the procedural period as a safeguard against the interests of stakeholders and a safeguard against academic integrity.

The project was also committed to ethical dilemmas by evaluating the need of adequate data and its protection, being reasonable in handling the expectations of the user and more so making the participation voluntary and free of any coercion. Using ethics as a part of all requirements engineering phases, the project issued professionally sound system specifications, but the stakeholders did not lose their trust. The last category of requirements provides a solid background of creation of a secure, efficient, scalable and friendly system of flight reservation that should fit the contemporary demands in traveling without overlooking the ethical principles and rights of stakeholders.

10.References

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11.Appendix

11.1 Individual Contribution Logs

Team Member	Role & Responsibilities	Key Deliverables
Nang Khin Myat Lay Phyu	1.Methodology 2. Requirement Specification 3.Traceability Matrix	- Methodology Section- Functional & Non-functional Requirements- Traceability Matrix
MD PARVEJ AHMED RAFI	1.Executive Summary- 2.Survey Design 3. Ethical Considerations	- Survey Design (Google Forms)- Executive Summary- Ethical Considerations Section
OMID QAZIKHIL	1.Introduction & Project Context 2.Reflection & Conclusion- 3.Change Management & Versioning	- Introduction & Project Context- Reflection & Conclusion- Change Management & Versioning

11.2 Survey/Interview Transcripts

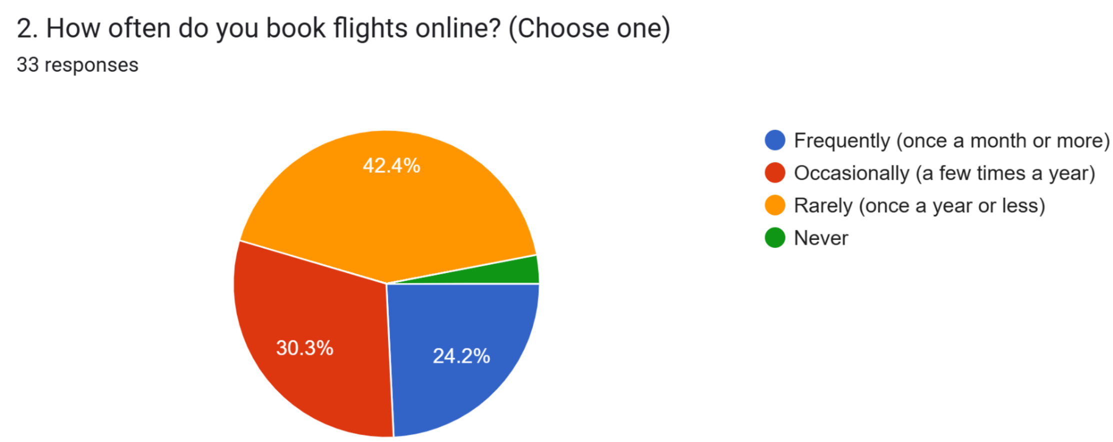

Figure 1

3. Which features do you expect from a flight reservation system? (Select all that apply)
33 responses

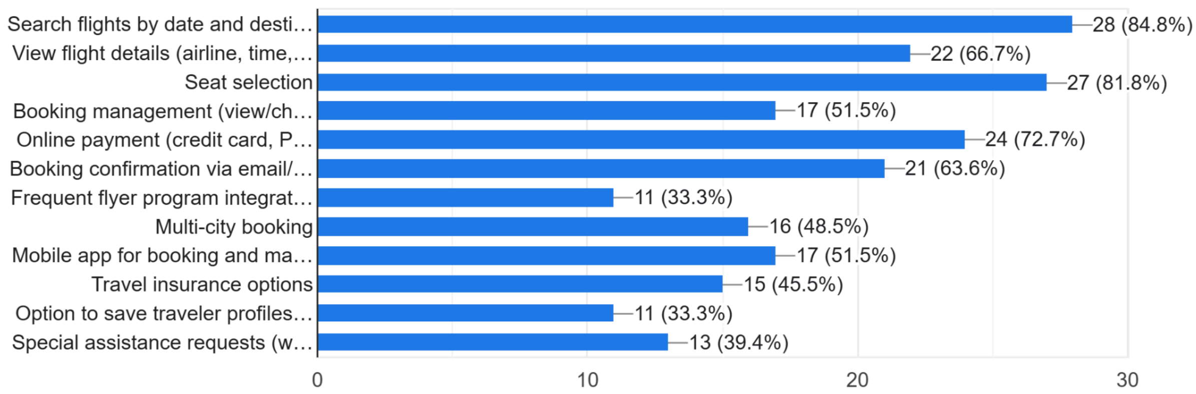

Figure 2

How important is real-time flight availability and pricing updates? (Rate from 1 to 5, where 1 is Not Important and 5 is Very Important)
33 responses

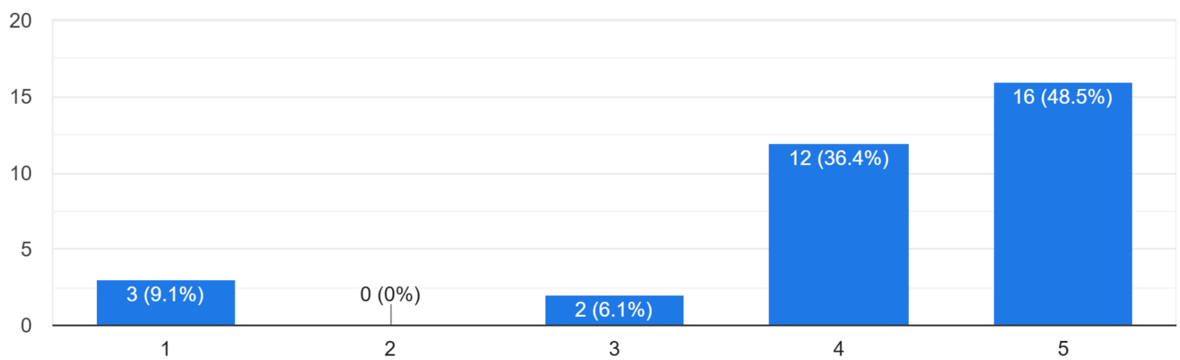

Figure 3

Should the system provide personalized recommendations based on user preferences or past bookings?

33 responses

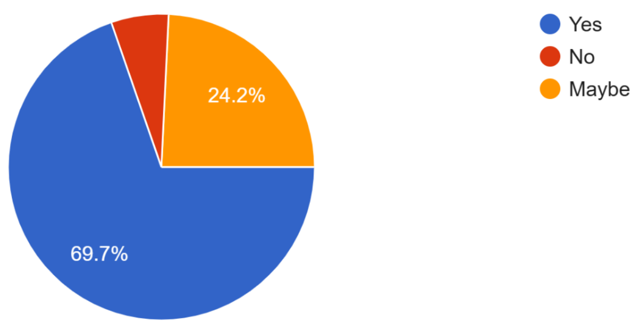

Figure 4

13. How important is multi-language support in the system? (Rate 1 to 5)

33 responses

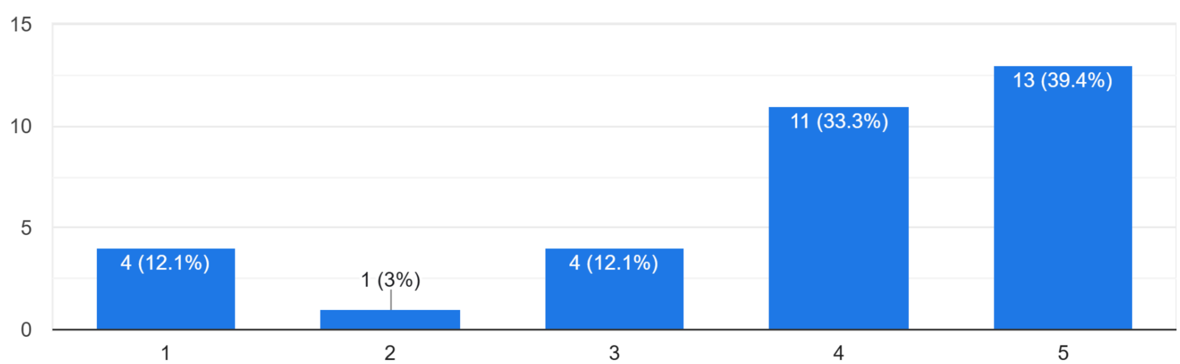

Figure 5

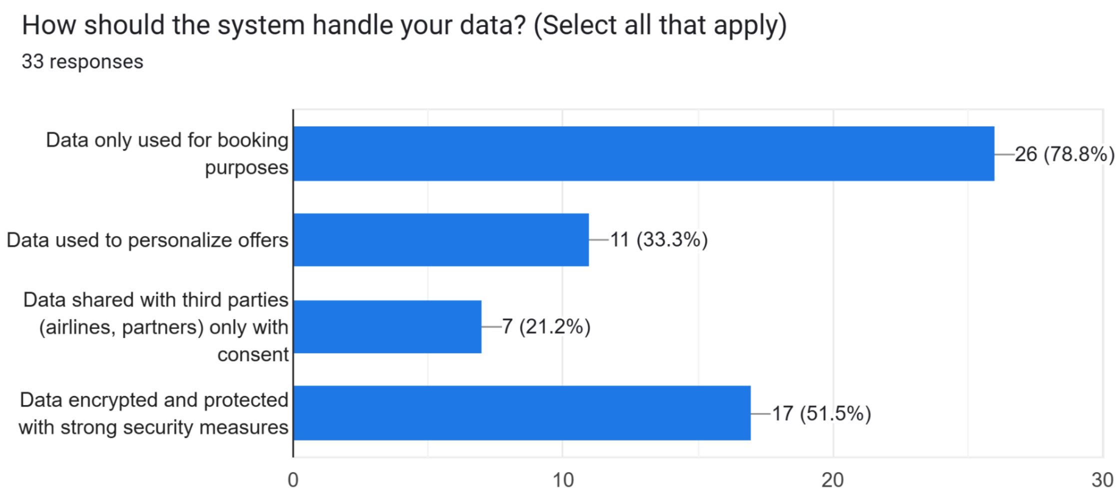

Figure 6

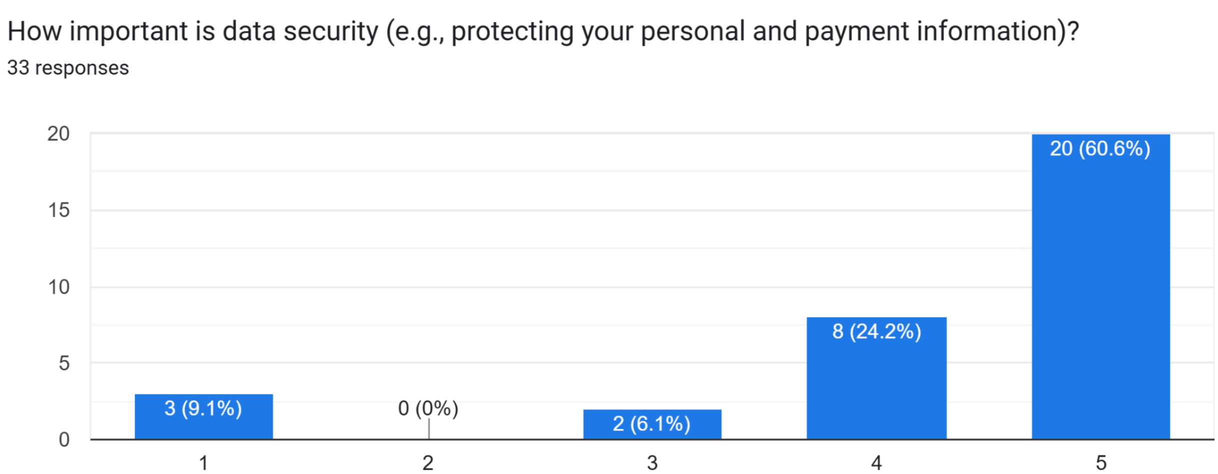

Figure 7

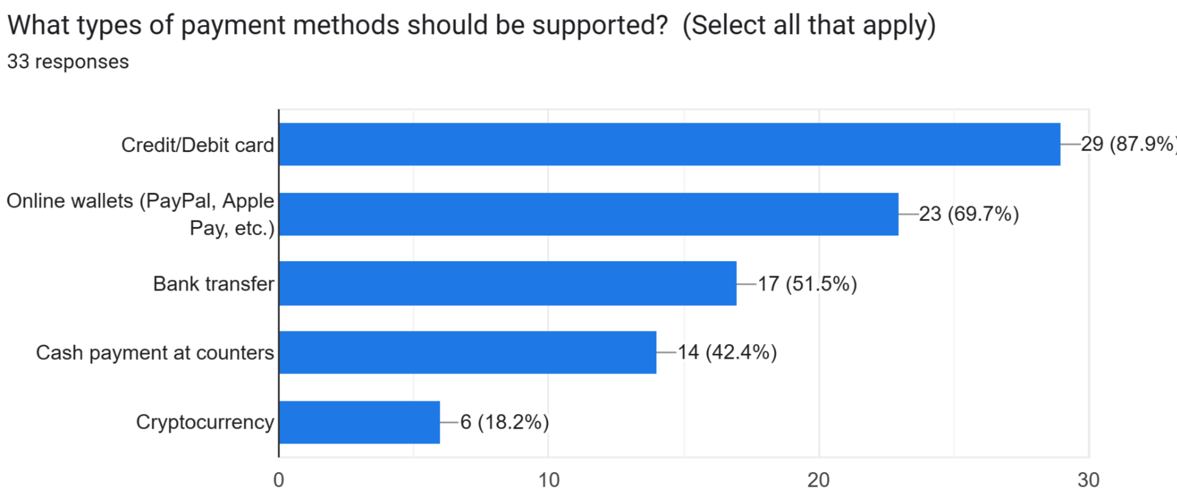

Figure 8

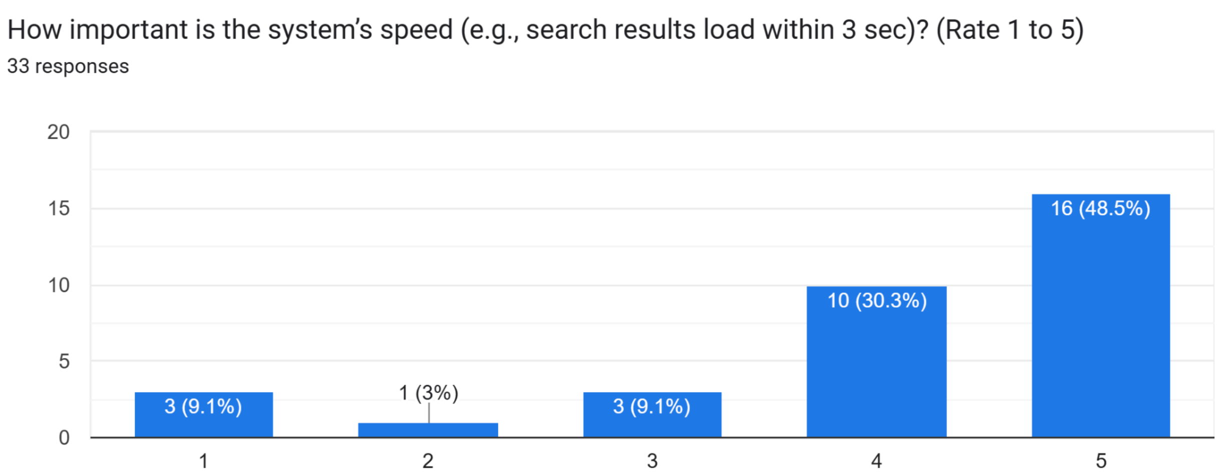

Figure 9

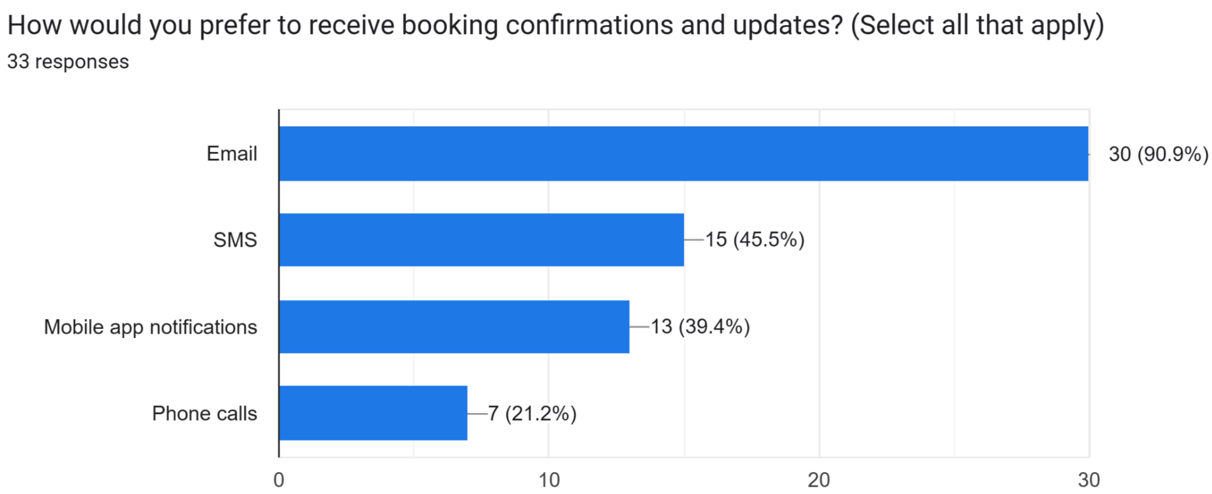

Figure 10

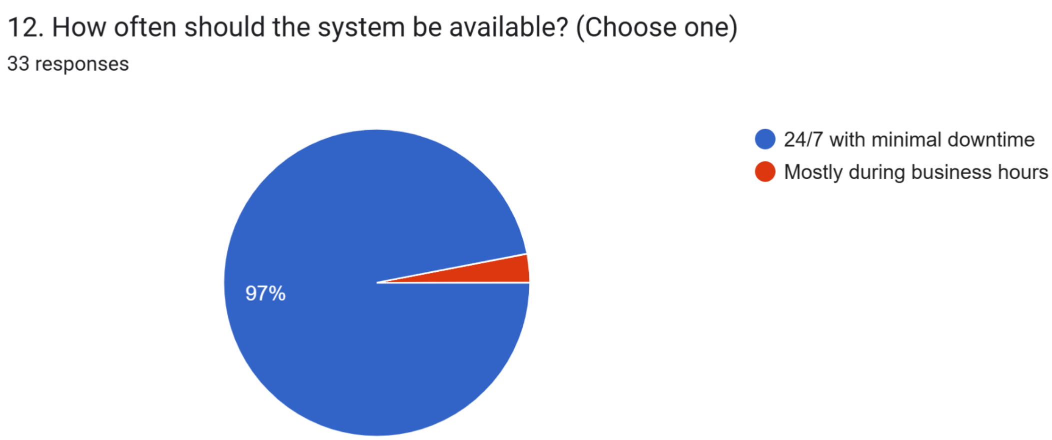

Figure 11

Are you comfortable with the system storing your personal and payment information for future bookings?

33 responses

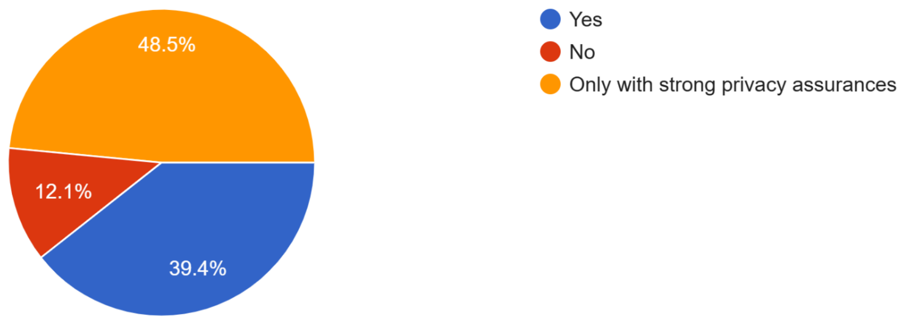

Figure 12

How should the system handle your data? (Select all that apply)

33 responses

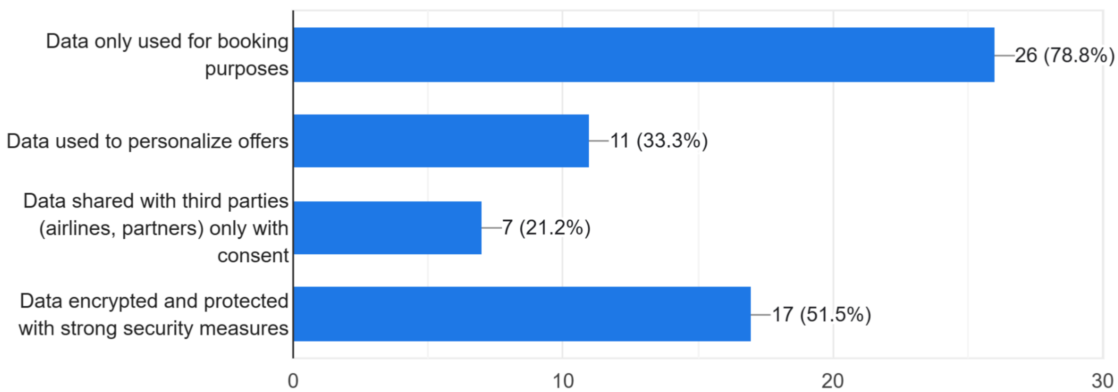

Figure 13

11.3 Relevant artefacts

Request for Input on Flight Reservation System Requirements Inbox ×

M

Md Parvej Ahmed Rafi <mdparvej.ahmedrafi@student.aiu.edu.my> 9:23 PM (0 minutes ago)
to airsomalia00

Hello ,

I hope this email finds you well.

We are currently working on the development of a Flight Reservation System as part of a requirements engineering project, aimed at improving efficiency, security, and user experience in online flight bookings. As part of our process, we are gathering input from key stakeholders in the flight industry to better understand the specific needs and expectations that should be addressed by the system.

We would appreciate it if you could take a moment to provide us with feedback on the following aspects:

1. **Core Functionalities**: What are the key features you would expect in a flight reservation system? (e.g., real-time flight search, seat selection, payment processing, etc.)
2. **User Roles**: What roles should be incorporated into the system? (e.g., passengers, travel agents, airline staff, administrators)
3. **Security & Privacy**: Are there any specific security or data privacy requirements that must be considered for protecting user information?
4. **Non-Functional Requirements**: Are there any system performance requirements such as speed, scalability, or availability that are particularly important to your operations?
5. **Additional Features**: Are there any other features or integrations (e.g., loyalty programs, dynamic pricing) that would add value to the system?

Your insights will be invaluable in shaping the design of the system, ensuring it aligns with industry standards and meets the needs of both passengers and airline staff.

Please feel free to provide any other thoughts or suggestions that you think are important. We are happy to schedule a call or meeting if you prefer a more detailed discussion.

Thank you for your time and assistance. We look forward to hearing from you.

Figure 14 : Email Request for Requirement